Code Sentinel

Project Plan

1. Overview and Tools Used

This document is the project plan for Code Sentinel, a multi-agent code review platform. The plan covers 17 weeks, from 27-Jul-2026 to 22-Nov-2026, and is divided into 9 work packages containing 35 tasks and 10 milestones. Every milestone is aligned with a course deliverable deadline. The team has 8 members and the planned effort is 200 person-hours per member, which is 1600 person-hours in total.

Client work covers the GitHub App and the Next.js dashboard.

Tool	**What it is used for**	**File in repo**
GanttProject 3.3	Task timeline, milestones, dependencies and per-member resource allocation.	CodeSentinel_ Project_Plan.gan
Trello	Kanban task board - 6 lists, 45 cards, one card per task with owner, labels and due date.	docs/charts/ trello_board.png
Git repository	Both files and this document are version controlled and submitted in the repo.	/docs folder

2. Minimum Viable Product (MVP)

The MVP for Code Sentinel is the API Gateway, the LangGraph Orchestrator, and two working agents (Style Agent and Security Agent), connected end-to-end so that a real pull request on GitHub triggers a review and a comment is posted back automatically.

- The MVP does not include the Logic, Performance or Documentation agents or the Next.js dashboard - these are built on top of the same architecture once the MVP is working.
- The MVP is done when: (a) a webhook from a real GitHub repo reaches the gateway, (b) the orchestrator fans out to both agents in parallel with timeout handling, (c) results are aggregated and de-duplicated, and (d) at least one suggested change is posted back to the pull request.
- This is tracked as milestone M5 (09-Oct, Week 11), an internal checkpoint ahead of the mid-semester demonstration (M6).

3. Project Timeline and Milestones

The 9 work packages run across the 17 weeks; milestones M1-M10 are listed below with the course deliverable each one satisfies. Work package 6 (Monitoring & Security) adds service metrics with Grafana dashboards and alerts, request-origin logging with a security audit trail, rate limiting, and a devices view where users can see and sign out their sessions.

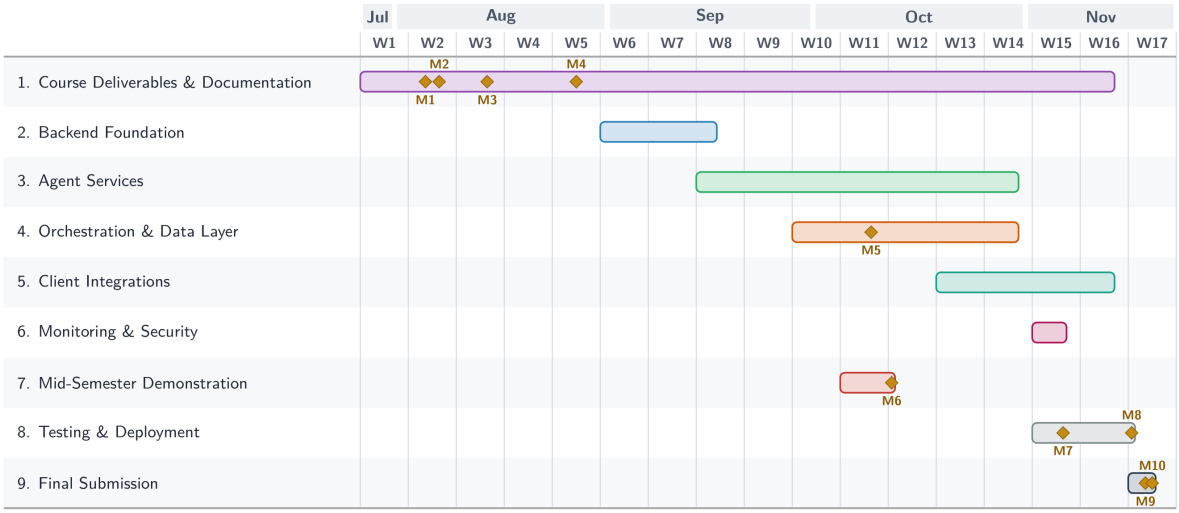
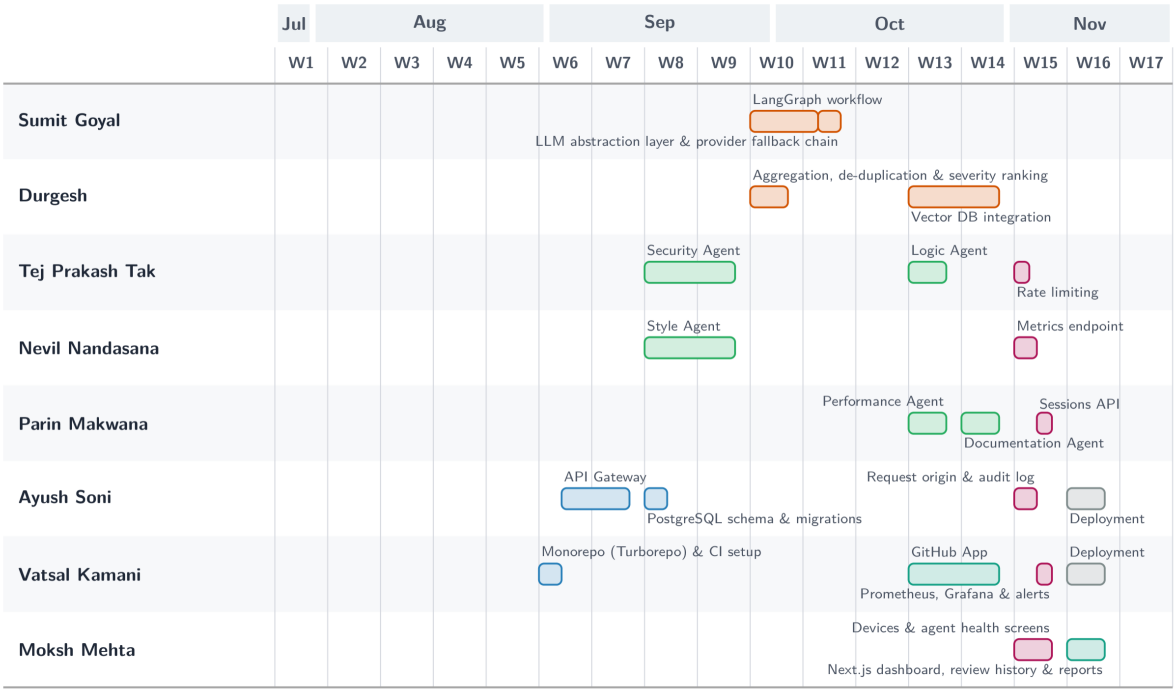


Figure 1: Work package timeline with milestones M1 to M10

ID	Milestone	Planned date	Week	Course deliverable it satisfies
M1	Project Proposal submitted (due 5-Aug)	05-Aug	W2	Deliverable 1 - Project Proposal
M2	Project Plan submitted (due 9-Aug)	07-Aug (due 9-Aug)	W2	Deliverable 2 - Project Plan
M3	Requirements Spec submitted (due 16-Aug)	14-Aug (due 16-Aug)	W3	Deliverable 3 - Requirements Spec (BRD)
M4	System Design Document submitted (27-Aug)	27-Aug	W5	Deliverable 4 - System Design Document
M5	Core pipeline working (gateway + orchestrator + 2 agents) - see Section 2	09-Oct	W11	Internal checkpoint (pre-demo)
M6	Mid-semester demonstration (week of 12-Oct)	12-Oct	W12	Deliverable 6 - Mid-Semester Demonstration
M7	Test Plan submitted (week of 2-Nov)	06-Nov	W15	Deliverable 7 - Test Plan & Test Cases
M8	Final application deployed, live URL (week of 16-Nov)	16-Nov	W17	Deliverables 8 & 9 - Final Application, Deployment
M9	Final report & presentation (18-Nov)	18-Nov	W17	Deliverables 10 & 11 - Final Report, Presentation
M10	Peer review submitted (19-Nov)	19-Nov	W17	Deliverable 12 - Peer Review

4. Individual Member Timelines and Task Assignments

Each member owns a specific module, and also takes a share of cross-team work - testing, integration and one supporting task on another member's module - so that ownership stays clear while contribution is spread evenly across the team. The tables below give the sub-tasks, dates and hours for each member.



Bars show each member's own module work. Shared course deliverables (documentation, testing, demo, final submission) involve all 8 members and are shown in Figure 1.

Figure 2: Individual timeline for each team member

4.1 Sumit Goyal - Orchestration

Sub-task	Dates	Hours	Definition of done
Design LangGraph state machine (nodes, edges, fan-out)	28-Sep - 30-Sep	20	Graph diagram approved and checked into /docs
Implement parallel fan-out to 5 agent services	01-Oct - 04-Oct	25	All 5 agents callable concurrently with mock responses
Implement per-agent timeout + partial-result handling	05-Oct - 06-Oct	15	Slow/failed agent does not block report generation
LLM abstraction layer (interface, prompt templates)	07-Oct - 08-Oct	15	Interface used by all 3 AI-suggestion agents
Provider fallback chain (OpenRouter to Gemini to Groq to Ollama)	09-Oct	10	Fallback triggers automatically on provider error
Unit tests for orchestrator (fan-out, timeout, fallback)	10-Oct - 11-Oct	10	Test suite covers all failure paths, runs in CI
Support task: LLM prompt tuning for Documentation Agent	28-Oct - 29-Oct	10	Docstring suggestions reviewed with Parin

Sumit Goyal: 105 hours.

4.2 Durgesh - Orchestration

Sub-task	Dates	Hours	Definition of done
Aggregation logic (merge results from 5 agents into one schema)	28-Sep - 29-Sep	15	Combined JSON report validated against schema
De-duplication (same issue flagged by 2+ agents)	30-Sep - 01-Oct	15	Duplicate rate under 5% on test diff set
Severity ranking / scoring model	02-Oct	10	Report sorted high to low severity, reviewed by team
Vector DB setup (Pinecone/ChromaDB) and embedding pipeline	19-Oct - 24-Oct	25	Code chunks embedded and stored on commit
Similarity search integration into orchestrator	27-Oct - 30-Oct	20	Past similar issues surfaced in report
Unit and integration tests for aggregation and vector search	31-Oct - 01-Nov	15	Test suite covers merge, de-dup and search paths
Support task: severity thresholds for Security Agent suggestions	02-Nov	5	Thresholds agreed with Tej and applied

Durgesh: 105 hours.

4.3 Tej Prakash Tak - Agent Services

Sub-task	Dates	Hours	Definition of done
Security Agent: SAST rule integration (Semgrep/Bandit)	14-Sep - 18-Sep	15	Flags 3 seeded vulnerable-code samples correctly
Security Agent: secret detection (regex + entropy check)	19-Sep - 25-Sep	15	Detects hardcoded API keys/passwords in test repo
Logic Agent: prompt design + edge-case detection logic	19-Oct - 21-Oct	15	Flags null-check / off-by-one bugs in sample diffs
Logic Agent: confidence scoring for AI suggestions	22-Oct - 23-Oct	10	Only high-confidence issues posted as suggestions
Unit tests for Security Agent and Logic Agent	24-Oct - 25-Oct	15	Test suite covers detection and scoring logic
GitHub App: end-to-end test on a live pull request	26-Oct - 30-Oct	15	Suggested change appears inline on a real test PR
Support task: Performance Agent test cases	31-Oct	5	Sample hot-path cases reviewed with Parin
Gateway rate limiting: per GitHub installation on webhooks, per user on the API, per IP on failed sign-in	02-Nov - 03-Nov	10	Over-limit requests get 429 with Retry-After; tests cover all three limits

Tej Prakash Tak: 100 hours.

4.4 Nevil Nandasana - Agent Services & Monitoring

Sub-task	Dates	Hours	Definition of done
Style Agent: language dispatcher (route by file extension)	14-Sep - 17-Sep	15	Routes .js/.ts to ESLint, .py to Ruff correctly
Style Agent: Docker sandbox for safe linter execution	18-Sep - 21-Sep	15	Linter runs in isolated container, no host access
Style Agent: ESLint/Prettier/Ruff/Black wiring + auto-fix	22-Sep - 25-Sep	15	Auto-fixable issues corrected without an LLM call
Service metrics: /metrics endpoint in the shared service kit - request rate, latency and errors, review-job duration and outcome, LLM calls and quota per provider	02-Nov - 04-Nov	15	Every service exposes /metrics and Prometheus scrapes all of them
Unit tests for Style Agent and service metrics	05-Nov - 06-Nov	15	Test suite covers the dispatcher, sandbox and metric counters
Support task: dashboard filter logic	07-Nov - 09-Nov	10	Repo/date/severity filters agreed with Moksh and wired
Bug-fixing rota (team-wide regression pass)	10-Nov - 11-Nov	5	Assigned bugs closed and retested

Nevil Nandasana: 90 hours.

4.5 Parin Makwana - Agent Services

Sub-task	Dates	Hours	Definition of done
Performance Agent: hot-path / complexity checks	19-Oct - 21-Oct	15	Flags O(n^2)+ loops and N+1 query patterns in samples
Performance Agent: suggestion generation + confidence score	22-Oct - 23-Oct	10	Suggestions posted only above confidence threshold
Documentation Agent: missing-docstring detection	26-Oct - 28-Oct	15	Flags public functions with no docstring
Documentation Agent: docstring suggestion generation	29-Oct - 30-Oct	15	Generated docstring accepted/edited by reviewer in demo
Unit tests for Performance Agent and Documentation Agent	31-Oct - 01-Nov	15	Test suite covers detection and generation logic
Support task: Next.js dashboard report detail view	02-Nov - 03-Nov	10	Detail view agreed with Moksh and wired to API
Bug-fixing rota (team-wide regression pass)	04-Nov	5	Assigned bugs closed and retested
Sessions API: list and sign out a user's devices (sessions with IP, device and last seen)	05-Nov - 06-Nov	10	Devices listed and revocable; every sign-out written to the audit log

Parin Makwana: 95 hours.

4.6 Ayush Soni - Backend & Database

Sub-task	Dates	Hours	Definition of done
API Gateway: project scaffold, auth middleware	03-Sep - 06-Sep	15	JWT-protected route returns 401 without token
API Gateway: routing to orchestrator + webhook receipt endpoint	07-Sep - 11-Sep	15	GitHub webhook payload accepted and forwarded
PostgreSQL schema design (reviews, users, installations)	14-Sep	5	ER diagram matches System Design Document
PostgreSQL migrations + seed scripts	15-Sep - 16-Sep	10	migrate up/down runs cleanly on a fresh DB
Request origin and audit log: client IP (behind a trusted proxy), session and user agent on every request log; audit events table	02-Nov - 04-Nov	15	Sign-ins, session changes and auth failures recorded with IP and request id
Deployment: Dockerfile + DB service wiring for production	09-Nov - 11-Nov	15	App boots against containerised Postgres in staging
Contract/integration tests across gateway, orchestrator and agents	12-Nov - 13-Nov	20	Postman/Newman collection passes in CI
Support task: DB seeding tests for review history	14-Nov	10	Dashboard review-history queries validated against seed data

Ayush Soni: 105 hours.

4.7 Vatsal Kamani - Integrations & DevOps

Sub-task	Dates	Hours	Definition of done
Monorepo (Turborepo) setup + shared configs	31-Aug - 01-Sep	10	All services build via a single turbo build
CI pipeline (GitHub Actions: lint, test, build on PR)	02-Sep	10	CI badge green on main; blocks merge on failure
GitHub App: webhook subscription + Octokit client setup	19-Oct - 23-Oct	15	App receives pull_request events from a test repo
GitHub App: Checks API + suggested-changes posting	24-Oct - 30-Oct	15	Suggested change appears inline on a real test PR
Monitoring stack: Prometheus, Grafana dashboards and alert rules (agent down, error rate, LLM quota near its limit)	05-Nov - 06-Nov	10	Grafana shows live data from every service; a test alert fires
Deployment: public hosting, live URL, CI/CD to production	09-Nov - 11-Nov	15	main deploys automatically; live URL reachable
End-to-end test: real GitHub PR through to posted comment	12-Nov - 13-Nov	15	Full pipeline verified against a live test repository
Support task: repo-level agent configuration in dashboard	14-Nov	10	Admin can enable/disable agents per repository, tested with Moksh

Vatsal Kamani: 100 hours.

4.8 Moksh Mehta - Frontend

Sub-task	Dates	Hours	Definition of done
Next.js dashboard: agent health and monitoring view (status, latency and version of each agent, link to Grafana)	02-Nov - 04-Nov	15	Every agent's live status shown; the view refreshes on its own
Next.js dashboard: devices screen (signed-in sessions with IP, device and last seen; sign out a device)	05-Nov - 06-Nov	10	User can see and revoke their own sessions
Next.js dashboard: review history list with filters (repo/date/severity)	09-Nov - 11-Nov	15	Past reviews listed and filterable
Next.js dashboard: report detail page and repo config screen	12-Nov - 13-Nov	15	Admin can open a report and enable/disable agents per repository
Documentation Agent: prompt design for docstring suggestions (with Parin)	26-Oct - 28-Oct	15	Generated docstrings reviewed against a sample function set
UI test pass for the dashboard	14-Nov	10	Manual test script covers every dashboard screen, signed off
Bug-fixing rota (team-wide regression pass)	15-Nov	5	Assigned bugs closed and retested

Moksh Mehta: 85 hours.

5. Testing

Testing runs continuously alongside development rather than only at the end. Each member writes unit tests for their own module as part of the sub-tasks in Section 4; the cross-cutting activities below are shared across the team.

Activity	Owner(s)	Tooling
Unit tests per module	Each member, for their own module (see Section 4)	Jest (Node), Pytest (Python)
Contract/integration tests across gateway, orchestrator and agents	Ayush, Durgesh	Postman/Newman collection, run in CI
End-to-end test: real GitHub PR through webhook to posted comment	Vatsal, Tej	Test GitHub repo + GitHub Actions
Dashboard UI test pass	Moksh, Nevil	Manual test script
Monitoring and security checks: metrics scraped, alerts fire, rate limits return 429, audit events written	Vatsal, Tej	Prometheus and Grafana, integration tests run in CI
Bug log triage, regression fixes and retest rota	Nevil, Parin, Moksh, plus rotating support from others	Trello In Review/Blocked lists, GitHub Issues

Activity	Owner(s)	Tooling
Test Plan & Test Cases document (Deliverable 7, milestone M7)	Whole team, coordinated by Ayush	Submitted 02-06 Nov

The Trello board tracks this through the In Review list, so a task only counts as done once a second member has checked it, in addition to the dedicated Testing & Deployment work package (02-Nov to 13-Nov).

6. Effort Distribution per Member

Each member's own module and testing hours are itemised in Section 4 (85-105 hours per member, including embedded unit-testing and one support task on another member's module). The balance of each member's 200 hours goes to the whole-team course deliverables listed under work package 1 in Section 3 - the proposal, this plan, the requirements spec, the system design document, weekly progress reports, the mid-semester demonstration, the final report and the peer review - plus the shared testing activities in Section 5.

Category	Hours per member
Own module work, including embedded unit tests and one support task (Section 4)	85-105
Shared testing activities (Section 5)	10-20
Whole-team course deliverables (Section 3, work package 1)	75-95
Total	200

7. Task Board (Trello)

The same 35 tasks and 10 milestones are tracked as 45 cards on a Trello board named Code-Sentinel. Every card is one row of the plan, so the board and the Gantt chart always show the same work. Cards move from left to right as the work progresses, and the two review columns are kept separate so that a task is only counted as done after another team member has checked it.

List	Cards	What it holds
Backlog	29	Work not yet started - everything from September onwards.
To Do	7	Next 30 days - requirements spec, system design, monorepo and API gateway.
In Progress	2	Project plan (this document and the Gantt chart), and the weekly progress reports.
In Review	0	Work finished and waiting for review by another team member.
Blocked	0	Work stopped because of a dependency or an unresolved problem.
Done	2	Project proposal document and milestone M1, submitted on 5-Aug.

As on 8-Aug-2026 the board shows 2 cards in Done (the project proposal and milestone M1), 2 cards in progress (this project plan and the weekly progress reports), 7 cards queued in To Do for the next 30 days, and 29 cards still in the backlog. The board is shared with all 8 team members, and each card holds the owner, the planned effort in working days, the tasks it blocks, and a definition of done. The Monitoring & Security cards are added to the Backlog.

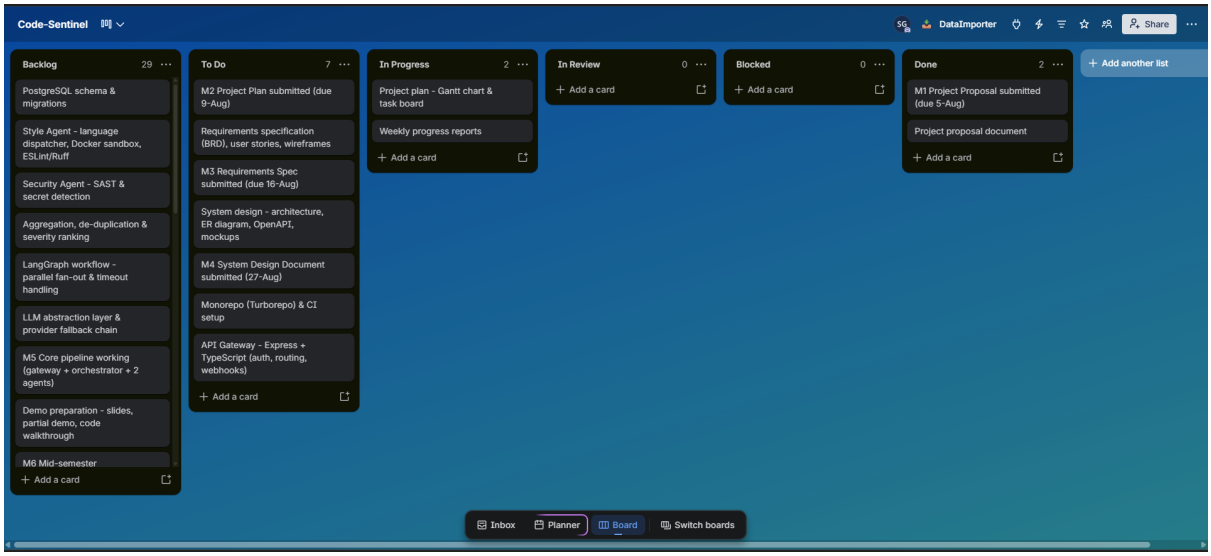


Figure 3: Trello task board as on 8-Aug-2026 (Deliverable 1 complete, project plan in progress)